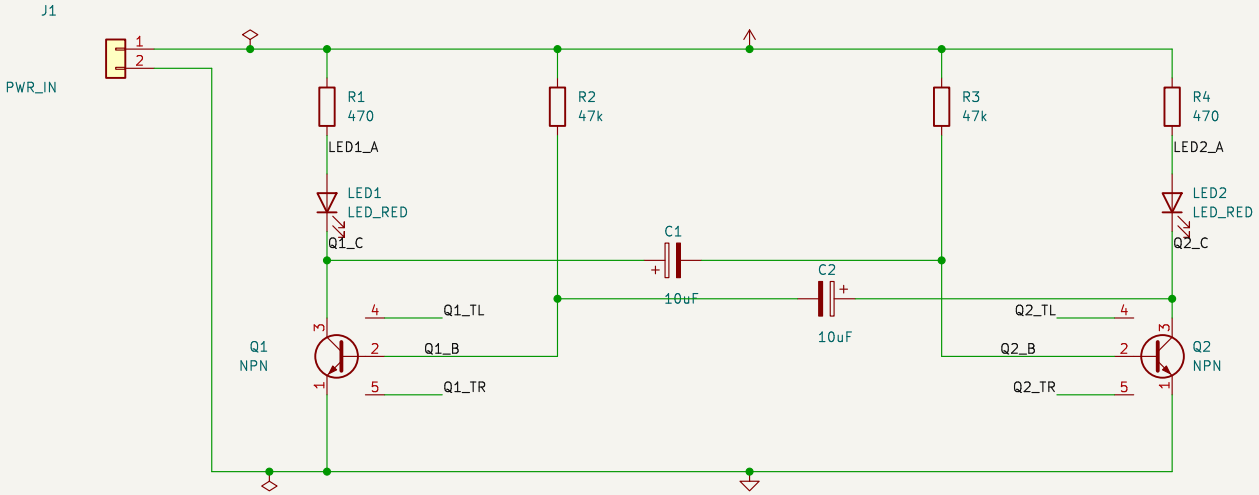
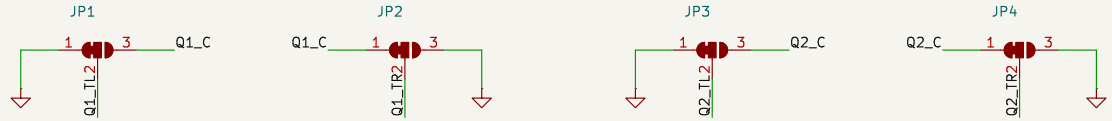


Pinout select (per transistor): pad 1 = GND (E), pad 2 = TO-92 outer leg, pad 3 = collector



Two-transistor astable multivibrator, LEDs flash alternately.
f = 1 / (1.386 * Rb * C) = 1.5 Hz with Rb = 47k, C = 10 uF. Supply 3 to 9 V (5 V recommended).
Every R / C / LED / Q position is a hybrid footprint: fit the THT part OR the 0805 / SOT-23 part.
JP1..JP4 select the TO-92 outer-leg pinout: bridged 1-2 (factory) = E-B-C (2N3904, PN2222, S8050);
cut 1-2 and bridge 2-3 = C-B-E (BC547, BC548, BC337). SOT-23 parts ignore the jumpers.

Hybrid THT / SMD footprints; TO-92 pinout selected by solder jumpers

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Title: Astable multivibrator LED flasher

Size: A4	Date: 2026-10-21	**Rev: 1.0**
KiCad E.D.A. 9.0.8		Id: 1/1